

- 10 The Norman army is advancing with constant speed u towards the Saxon army, which is at rest. When the armies are d apart, a Saxon horseman rides from the Saxon army directly towards the Norman army at constant speed x . Simultaneously a Norman horseman rides from the Norman army directly towards the Saxon army at constant speed y , where $y > u$. The horsemen ride their horses so that $y - 2x < u < 2y - x$.

When each horseman reaches the opposing army, he immediately rides straight back to his own army without changing his speed. Represent this information on a displacement-time graph, and show that the two horsemen pass each other at distances

$$\frac{xd}{x+y} \quad \text{and} \quad \frac{xd(2y-x-u)}{(u+x)(x+y)}$$

from the Saxon army.

Explain briefly what will happen in the cases (i) $u > 2y - x$ and (ii) $u < y - 2x$.